(d) Cosmology
Students should:
8.13P describe the past evolution of the universe and the main arguments in favour of the Big Bang theory
8.14P describe evidence that supports the Big Bang theory (red-shift and cosmic microwave background - CMB - radiation)
8.15P describe that if a wave source is moving relative to an observer, there will be a change in the observed frequency and wavelength
8.16P use the equation relating to change in wavelength, reference wavelength, velocity of a galaxy and the speed of light: $\frac{\text{change in wavelength}}{\text{reference wavelength}} = \frac{\text{velocity of a galaxy}}{\text{speed of light}}$ $\frac{\lambda - \lambda_0}{\lambda_0} = \frac{\Delta\lambda}{\lambda_0} = \frac{v}{c}$
8.17P describe the red-shift in light received from galaxies at different distances away from the Earth
8.18P explain why the red-shift of galaxies provides evidence for the expansion of the universe